

Experiment 06: Input & Output Manual

Design a Web Page using Node.js (Pure Core Modules)

🎓 Student & Laboratory Credentials

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- **Class / Division:** B.Tech Computer Engineering (Sem VI)
- **Academic Year:** 2026 – 2027
- **Course:** Skill Based Lab - Advanced Web Technology (SBL-AWT)
- **Institution:** Ramrao Adik Institute of Technology (RAIT), D Y Patil Deemed to be University, Nerul, Navi Mumbai
- **Problem Statement Domain:** HealthPulse — Digital Healthcare & Clinical Management Portal
- **Experiment Title:** Design a Web Page using Node.js

1. Experiment Aim

To develop a high-performance, lightweight HTTP web server using native **Node.js core modules** (`http` , `url` , `os` , `path` , `fs`) with zero external framework dependencies, demonstrating low-level request routing, streaming chunk aggregation, server diagnostics, and dynamic HTML templating.

2. 🖱️ Input Specification

2.1 File System Input

File Name	Format	Role & Implementation Responsibility
<code>server.js</code>	Pure Node.js (CommonJS)	Uses <code>http.createServer()</code> , custom router, <code>req.on('data')</code> / <code>req.on('end')</code> binary buffer aggregator, OS telemetry injection (<code>os.freemem</code> , <code>os.totalmem</code>), and dynamic HTML view generator.
<code>package.json</code>	JSON	Project metadata and start script (<code>node server.js</code>). Has zero dependencies (<code>dependencies: {}</code>).

2.2 Execution Command Input

```
# Navigate to the experiment directory
cd /home/ubuntu/SBL-EXPT/Asmit_25CE1051/Expt-06-NodeJS-WebPage

# Launch web server using pure Node.js (Default Port: 3000)
node server.js

# Or specify a custom port:
PORT=3555 node server.js
```

2.3 HTTP Route Dispatcher Input Mapping

HTTP VERB	PATHNAME	CONTENT-TYPE RETURNED	LOGICAL RESPONSIBILITY
GET	/	text/html; charset=utf-8	HealthPulse Portal Home + Metrics
GET	/about	text/html; charset=utf-8	Clinical specialties & hospital info
GET	/consultation	text/html; charset=utf-8	Interactive consultation intake form
POST	/consultation	text/html; charset=utf-8	Stream chunk parser & ticket output
GET	/api/status	application/json	System diagnostics & memory usage
ANY	(Unmatched paths)	text/html; charset=utf-8	Custom HTTP 404 Error page

2.4 Sample Input Payloads & Test Requests

Request 1: Server Diagnostics Query (`GET /api/status`)

```
curl -i http://localhost:3555/api/status
```

Request 2: Consultation Booking via Stream Body (`POST /consultation`)

```
curl -X POST http://localhost:3555/consultation \
  -H "Content-Type: application/x-www-form-urlencoded" \
  -d "fullName=Asmit+Jogdand&email=asmit.jogdand%40healthpulse.org&phone=9876543210&department=Cardiology&symptoms=Routine+EHR+Cardiac+Screening"
```

3. Output Specification

3.1 Terminal & Server Execution Logs

```
$ PORT=3555 node server.js
[HealthPulse] Native Node.js server is running at http://localhost:3555
[HealthPulse] Student: Asmit Jogdand (25CE1051)
[9:30:43 AM] HTTP GET request to: /api/status
[9:30:43 AM] HTTP POST request to: /consultation
[Stream Parser] Received 142 bytes of chunked payload. Parsing URL-encoded parameters.
[HealthPulse] Generated Clinical Consultation Ticket: HP-253968 for patient Asmit Jogdand.
```

3.2 HTTP REST API Diagnostics Output (`GET /api/status`)

```
HTTP/1.1 200 OK
Content-Type: application/json
Date: Thu, 01 Oct 2026 09:30:43 GMT
Connection: close

{
  "status": "ONLINE",
  "application": "HealthPulse Node Server",
  "developer": "Asmit Jogdand (25CE1051)",
  "college": "RAIT, D Y Patil Deemed to be University",
  "nodeVersion": "v24.20.0",
  "memory": {
    "free": "4504.99 MB",
    "total": "11927.42 MB"
  },
  "uptimeSeconds": 42,
  "timestamp": "2026-10-01T09:30:43.868Z"
}
```

3.3 Dynamic Consultation Confirmation Receipt Output ([POST /consultation](#))

Upon receiving and parsing the streamed chunks, the server returns an HTML response containing the dynamic confirmation ticket:

```
+-----+
|                                     |
|               HEALTHPULSE TELEHEALTH CONSULTATION TICKET               |
|                                     |
| Booking Reference Token: HP-253968                                     |
| Server Timestamp:      2026-10-01 09:30:43 IST                         |
|                                     |
| Patient Details:                                                       |
| • Full Legal Name:      Asmit Jogdand                                 |
| • Communication Email:  asmit.jogdand@healthpulse.org                 |
| • Cellular Contact:     +91 9876543210                               |
| • Clinical Specialty:   Cardiology & CathLab                         |
| • Chief Symptoms:      Routine EHR Cardiac Screening                 |
|                                     |
| Telemetry Diagnostics:                                                 |
| • Served by Node.js Core: v24.20.0                                   |
| • Memory Consumption:   Active RSS 34.2 MB                           |
|                                     |
| [ Print Consultation Pass ]      [ Return to Portal Home ]             |
|                                     |
+-----+
```

3.4 Graphical User Interface Output Representation ([GET /](#))

```
+-----+
| [HealthPulse Portal]  SBL - Advanced Web Technology (Sem VI)           | Student: Asmit (25CE1051) |
+-----+
| (♥ HealthPulse Telehealth)  Home      About Clinical Services          | Book Consultation        |
+-----+
|                                     |
|               HEALTHPULSE TELEHEALTH NODE CORE ENGINE               |
|               High-Speed Native HTTP Server Built Using Pure Node.js Modules               |
|                                     |
| --- Real-Time Server Performance & OS Diagnostics ---+-----+
| | Node Engine: v24.20.0      | Server Architecture: x64 Linux      | Process Uptime: 42s      |
| | Available RAM: 4,504.99 MB / 11,927.42 MB Total Memory      | Host OS: Linux Ubuntu    |
| +-----+
|
| --- Instant Consultation Request ---+-----+
| | Full Name: [ Asmit Jogdand      ] Email Address: [ asmit@healthpulse.org      ] | |
| | Phone No:  [ 9876543210        ] Specialty:    [ Cardiology & CathLab      v] |
| | Symptoms:  [ Routine cardiac assessment and blood pressure evaluation          ] |
| |                                     |                                     |
| |               [ Dispatch Consultation Request (POST) ]               |
| +-----+
+-----+
```

3.5 Test Case Execution Results

Test Scenario	Injected Request	Native Node.js Execution Handling	Observed Response Output	Status
Home Page Route	GET /	Matches / in routing switch	HTTP 200; renders HTML with live OS memory metrics	PASS
About Page Route	GET /about	Matches /about	HTTP 200; renders hospital department overview	PASS
Metrics Endpoint	GET /api/status	Sets Content-Type: application/json	HTTP 200; JSON payload with free/total memory and uptime	PASS
Stream Chunk POST	POST /consultation	Gathers chunk buffers, parses form body	HTTP 200; dynamic confirmation pass with ticket HP-253968	PASS
404 Invalid Path	GET /nonexistent	Hits default fallback case	HTTP 404 Not Found; displays customized error page	PASS

Test Scenario	Injected Request	Native Node.js Execution Handling	Observed Response Output	Status
Zero Dependencies	Clean environment check	Checks <code>package.json</code> dependencies	Runs purely on standard Node.js library (<code>http</code> , <code>os</code> , <code>url</code>)	PASS

4. Conclusion & Verification

Experiment 06 successfully demonstrated native Node.js HTTP server architecture. The server handled routing, chunked data streams, process telemetry, and dynamic HTML rendering without relying on third-party frameworks like Express, confirming fundamental mastery of the Node.js event-driven runtime.